

Capture-the-Flag Challenges in the Age of Generative AI

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Capture-the-Flag (CTF) challenges are hacking tasks aiming to build and test cybersecurity skills. They rely on ingenuity and technical understanding, with difficulty levels ranging from beginner to expert. They are also a tool commonly used in cybersecurity education, as they allow a certain level of gamification of classes.

Generative AI, specifically large language models, can be leveraged to assist in solving such challenges, which shift the required skill set as knowledge of technical details becomes less important and AI-related skills like finding good prompts can possibly make up for in-depth technical knowledge. In this project, it is your task to test to what extent traditional CTF challenges can be solved by AI and to develop challenges that are suitable for the current state of freely available generative AI models.

1 Project Description and Steps

Each student is assigned a category of challenges, which the categories being listed at the end of this section.

After you have selected/are assigned a category, you first find challenges on the web. You should find **at least 1** challenge for each of the following expertise levels:

- Interested high-school student/first-year Bachelor student
- CS Bachelor student in at least their second year without any specific background in security
- CS Bachelor/Master student with some security knowledge (e.g., one course)
- CS Master student specializing in security
- Professional security expert

You should then use various freely available AI models to solve the challenge. For each challenge, consider two scenarios:

1. **Fully AI:** Giving the challenge to the AI as a direct prompt without any modifications
2. **Assisted AI:** Use the AI as an assistant to solve the challenge, e.g., by asking for information, letting it write code, using it to iterate over a number of possibilities

Document the prompts you used as well as the results.

Afterwards, you should come up with a list of challenges (again at least one per expertise level) that still require significant user involvement to be solved, i.e., in the assisted AI scenario, the main ideas should still come from the human or the AI fails. It is possible to use some of your original challenges here if they were found to be hard for an AI.

At the end of the project, you should give both sets of challenges to the other people working on CTF to solve and compare their results with your own. They will be allowed to use AI to solve the challenges.

Categories of Challenges:

- Cryptanalysis (i.e., breaking cryptographic protocols)
- Network traffic analysis and crafting of specific traffic
- Database exploits
- Web exploits
- Binary exploitation (e.g., buffer overflows)
- Forensics (i.e., find hidden data and information in data)
- Reverse Engineering (i.e., learn source code of closed-source product)

I didn't include categories like hardware or lock picking, as they require additional equipment. If someone wants to work on another category and can argue that the category does not significantly overlap with the ones above, we can discuss whether the option is viable.

2 Resources

CTFs with example challenges:

- picoCTF: <https://picoctf.org/>
- HackerDNA: <https://hackerdna.com/labs>
- OvertheWire: <https://overthewire.org/wargames/>
- DEFCON CTF: <https://defcon.org/html/links/dc-ctf.html>

There are more CTFs available on the web, these are just starting points.

3 Milestones/Deadline

1. May 15: set of initial challenges
2. June 1: performance of initial challenges when AI models are used
3. July 1: modified challenges and their performance when AI models are used (**HARD**)
4. July 15: Solve challenges of other students and provide your solution and feedback to them (**HARD**)
5. September 15: Submit deliverables (**HARD**)

Hard deadlines mean that if you don't meet the deadline without a good reason, you can't continue the project. The only hard deadlines are deadlines related to the peer feedback and the final deadline. If you miss any other deadlines, you won't get feedback, which may lead to failing the project as you might not realise any mistakes you make.

4 Deliverables

At the end of the project, you should submit the following documents:

1. List of initial challenges selected, including the assigned expertise level and an explanation why you gave the level (in pdf format, potentially with a .zip for data necessary to solve the challenges)
2. List of modified challenges selected, including the assigned expertise level and an explanation why you gave the level (in pdf format, potentially with a .zip for data necessary to solve the challenges)
3. A report on how AI performed on the original challenges, including which AI models you leveraged and a list of prompts and answers, as well as information on how your fellow students performed (in pdf)
4. A report on how AI performed on the modified challenges, including which AI models you leveraged and a list of prompts and answers, as well as information on how your fellow students performed (in pdf)
5. Sample solutions for all challenges (format depends on challenge)
6. AI statement: description on how you used AI in the project (pdf, only needs to contain the parts not in the other reports, e.g., use of AI to improve writing or to find initial challenges)